SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

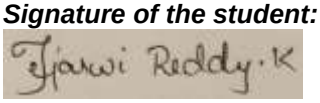
Annexure A

ARTICLE REVIEW

Code of Conduct:

I Tejaswi Reddy K (192371036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RESEARCH ARTICLE REVIEW

Selected Article

Ismail, L., Materwala, H., Tayefi, M., Ngo, P., & Karduck, A. P. (2022).
"Type 2 Diabetes with Artificial Intelligence Machine Learning: Methods and Evaluation."
Archives of Computational Methods in Engineering, 29, 313–333.
DOI: 10.1007/s11831-021-09582-x

The article was published online by Springer on 15 April 2021 and appeared in Volume 29 of the journal in 2022. It is an open-access original research paper.

Publisher / Article: Springer Nature – Full Article and PDF

DOI: 10.1007/s11831-021-09582-x

TASK 1: ARTICLE SUMMARY

1(a) Citation, Domain and ML Problem

Full Citation

Ismail, L., Materwala, H., Tayefi, M., Ngo, P., & Karduck, A. P. (2022). *Type 2 Diabetes with Artificial Intelligence Machine Learning: Methods and Evaluation*. Archives of Computational Methods in Engineering, 29, 313–333. https://doi.org/10.1007/s11831-021-09582-x.

Domain

The research belongs to:

Healthcare / Medical Diagnosis / Diabetes Risk Prediction

The specific application is **prediction of Type 2 diabetes using machine-learning classification algorithms**.

ML Problem

The problem is a **binary classification problem**:

- Class 0 → Non-diabetic
- Class 1 → Diabetic

The researchers investigate whether different machine-learning algorithms can accurately distinguish diabetic from non-diabetic individuals using demographic, medical, hereditary and lifestyle-related risk factors.

The study evaluates **35 machine-learning algorithms**, both with and without feature selection, across three real-world diabetes datasets.

1(b) Key Objective and Research Gap

Article Summary — approximately 180 words

The main objective of the article is to conduct a unified comparison of machine-learning algorithms for Type 2 diabetes prediction. The authors observed that previous diabetes-prediction studies used different datasets, preprocessing methods and evaluation metrics, making it difficult to determine which algorithms actually perform better. The research gap was therefore the absence of an objective and consistent comparison across multiple real-world diabetes datasets. To address this gap, the authors evaluated 35 machine-learning algorithms using the PIMA Indian, UCI diabetes and MIMIC III datasets. They also investigated feature-selection techniques to determine which diabetes risk factors contribute most strongly to prediction. The models were evaluated using accuracy, F-measure and execution time under a common

experimental setup. Their results showed that model performance is highly dependent on the characteristics of the dataset. Bagging combined with Logistic Regression performed best on balanced data, whereas Random Forest performed better on highly imbalanced datasets. The authors also emphasized that accuracy alone can be misleading in healthcare and recommended using F-measure and feature selection when evaluating diabetes prediction systems.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques and Datasets

The researchers implemented their experiments using **Weka 3.8**.

They evaluated a large collection of algorithms, including:

- Decision Tree
- Naïve Bayes
- Bayesian Network
- Logistic Regression
- Support Vector Machine
- K-Nearest Neighbours
- Random Forest
- Random Tree
- REPTree
- Artificial Neural Network
- K-means combined with classifiers
- Bagging
- Boosting
- Stacking
- OneR
- ZeroR
- JRip
- Decision Table

In total, **35 algorithms** were investigated.

Decision Tree

The Decision Tree selects a feature that gives the highest information gain at each node.

The information gain is represented as:

[IG(R)=H(D)-H(D|R)]

where:

- (H(D)) = entropy of diabetes classes
- (H(D|R)) = entropy after splitting using risk factor (R)

Thus, the feature that produces the greatest reduction in uncertainty is selected for the next decision node.

Logistic Regression

Logistic Regression estimates the probability that an individual belongs to the diabetic class:

[P(Diabetes)=\frac{e^{\beta_0+\sum\beta_iR_i}}{1+e^{\beta_0+\sum\beta_iR_i}}]

The coefficients represent the contribution of the risk factors to the predicted probability.

Naïve Bayes

Naïve Bayes calculates the probability of a class given the observed risk factors using Bayes' theorem and assumes conditional independence between features given the class.

Datasets

The paper evaluates three datasets:

Dataset	Features	Observations	Diabetic	Non-diabetic
PIMA Indian	9	537	179	358
UCI	12	65,840	51,034	14,806
MIMIC III	4	39,698	1,242	38,456

The preprocessed PIMA dataset contains pregnancy count, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age and the diabetes outcome. The UCI dataset contains demographic and medical factors, while MIMIC III contains ethnicity, gender, age and diabetes information.

Data Preparation

The researchers:

1. Prepared the three datasets.
2. Converted relevant diagnosis information into binary risk-factor variables.
3. Created diabetes class labels.
4. Investigated feature selection.
5. Used **10-fold cross-validation**.
6. Repeated each algorithm three times and averaged accuracy, F-measure and execution time.

2(b) Mapping to CO-4 Topics

CO-4 Topic	Application in Article
Inductive Learning	Models learn general diabetes-prediction rules from labelled patient records.
Decision Trees	Decision Tree, Random Tree and REPTree algorithms are evaluated.
Statistical Learning	Logistic Regression, Naïve Bayes and Bayesian Network are evaluated.
Classification	Diabetes/non-diabetes is treated as a binary classification problem.
Feature Selection	Nine feature-selection algorithms are used to identify important predictors.

Methodological Strength

A major strength is the **unified experimental setup**.

Instead of comparing individual papers that use different datasets and metrics, the researchers evaluated 35 algorithms using the same three datasets and a consistent evaluation procedure. This makes the comparison more meaningful.

Methodological Limitation

A major limitation is that the datasets have very different characteristics and substantial class imbalance. In particular, MIMIC III contains only **1,242 diabetic observations compared with 38,456 non-diabetic observations**. This can make accuracy misleading and can cause models to favor the majority class.

Another limitation is that the study primarily evaluates predictive performance rather than demonstrating prospective clinical deployment.

TASK 3: RESULTS & CRITICAL EVALUATION

3(a) Key Results

The researchers found that **performance depended strongly on the dataset**.

Main findings

Dataset Type	Best-performing approach reported	Main observation
PIMA Indian	K-means + Logistic Regression / Bagging-LR	Logistic Regression-based approaches performed strongly
Balanced datasets	Bagging + Logistic Regression	Reported as the most accurate
UCI	Random Forest	Strong performance on imbalanced data
MIMIC III	Random Forest	Better ability to identify minority-class cases
Feature-selected datasets	Similar relative ranking	Reduced feature count also reduced execution time

The authors report that **Bagging-LR was the most accurate for balanced datasets**, while **Random Forest was the most accurate for imbalanced datasets**.

The authors also found that some algorithms produced high accuracy while failing to predict the minority diabetic class. This is an important finding for healthcare applications because a model that simply predicts the majority class can achieve apparently high accuracy while missing diabetic patients.

Important note about reported metrics

The article presents many of the algorithm-level accuracy and F-measure results graphically rather than as a complete numerical table. Therefore, it would be academically unsafe to invent exact percentages by reading approximate values from the figures.

The paper explicitly reports the **relative winners and experimental conclusions**, which are more reliable for this review:

- Bagging-LR → strongest on balanced data.
- Random Forest → strongest on imbalanced data.
- Accuracy alone can be misleading.
- F-measure is important for minority-class detection.
- Feature selection reduces execution time without necessarily degrading performance.

Execution-Time Examples

The paper provides numerical execution times. With feature selection, for example:

Algorithm	PIMA	UCI	MIMIC III
Decision Tree	~0 min	0.216 min	0.016 min
Naïve Bayes	~0 min	0.016 min	~0 min
Logistic Regression	~0 min	0.116 min	0.234 min
Random Forest	0.016 min	2.116 min	0.600 min
SVM-RBF	0.016 min	574.616 min	12.510 min
ANN	0.033 min	1449.850 min	46.600 min

These values demonstrate the computational advantage of simpler statistical/tree models compared with computationally expensive models on larger datasets.

3(b) Critical Evaluation

Are the Results Realistic?

The results are generally plausible because the study uses real-world datasets and compares a broad range of algorithms.

However, extremely high accuracy in medical datasets should always be interpreted carefully.

The authors themselves demonstrate why:

Suppose:

[\text{Non-diabetic}=97%] and [\text{Diabetic}=3%]

A classifier that predicts every patient as non-diabetic could achieve approximately 97% accuracy while having **zero useful diabetic detection capability**.

The paper identifies exactly this issue in the UCI and MIMIC III experiments: several models favored the majority class and therefore failed to produce meaningful F-measure values for the minority class.

Potential Dataset Bias

The three datasets originate from different populations and healthcare settings.

Therefore:

- demographic distributions differ,
- feature availability differs,
- disease prevalence differs,
- healthcare practices differ.

A model trained on one population may not perform equally well on another.

Evaluation Bias

The authors use 10-fold cross-validation, which is stronger than a single train/test split. However, cross-validation within the same dataset does not necessarily demonstrate performance on a completely independent hospital or geographical population.

Additional Experiments That Would Strengthen the Research

I would add:

1. **External validation** using an independent hospital dataset.
2. **Temporal validation** — train on earlier patients and test on later patients.
3. **Calibration analysis** to determine whether predicted probabilities correspond to actual diabetes risk.
4. **ROC-AUC and PR-AUC** rather than relying mainly on accuracy/F-measure.
5. **Sensitivity and specificity** because false negatives are clinically important.
6. **Fairness analysis** across age, sex and ethnicity.
7. **Confidence intervals** around performance metrics.
8. **Prospective clinical validation** before deployment.

3(c) Real-World Applicability

The techniques could be deployed as **clinical decision-support systems**.

Possible applications

1. Primary-care screening

Patient information could be entered into a system that calculates diabetes risk.

2. Hospital decision support

An ML model could flag high-risk patients for additional clinical evaluation.

3. Preventive healthcare

High-risk patients could be referred for lifestyle interventions.

4. Population health

Healthcare organizations could identify groups requiring targeted diabetes screening.

5. Electronic Health Records

The model could operate on structured EHR data and provide a risk score to healthcare professionals.

Industry Scaling Challenges

Data availability

Large healthcare datasets may contain:

- missing values,
- inconsistent formats,
- duplicated records,
- measurement errors,
- different coding systems.

Data imbalance

Diabetic and non-diabetic populations may be highly imbalanced, making accuracy unreliable.

Privacy

Patient information is sensitive and must be protected through appropriate privacy, security and governance mechanisms.

Model generalization

A model trained using one hospital's population may not perform equally well at another hospital.

Explainability

Healthcare professionals need to understand why the system generated a prediction.

Ethical concerns

Incorrect predictions can cause:

- missed diagnoses,
- unnecessary follow-up,
- unequal performance between patient groups,
- loss of trust.

Therefore, the model should support—not replace—clinical judgement.

TASK 4: REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

Insight 1: Accuracy is not always sufficient

The article demonstrates that a model can obtain high accuracy simply by predicting the majority class.

This deepened my understanding of **classification evaluation and inductive learning**. In a healthcare problem, the learner must generalize correctly to both classes rather than simply maximize overall accuracy.

CO-4 connection: Inductive Learning + Classification Evaluation.

Insight 2: Decision Trees select features using information gain

The article explains that a Decision Tree chooses a feature that maximizes information gain.

[IG=H(D)-H(D|R)]

This helped me understand that the root node is not selected randomly. It is selected because the corresponding feature provides the greatest reduction in uncertainty.

CO-4 connection: Decision Trees + Information Gain.

Insight 3: Model performance depends on the dataset

The most important insight is that there is no universally best ML algorithm.

The study found that Bagging-LR was strongest on balanced data, while Random Forest was more effective for imbalanced datasets.

This demonstrates that algorithm selection should consider:

- data size,
- class distribution,
- feature characteristics,
- computational requirements,
- application requirements.

CO-4 connection: Statistical Learning + Model Selection.

4(b) Proposed Novel Extension

Proposed Experiment: Explainable and Fair Diabetes Risk Prediction Using External Validation

I would extend the research by developing a **clinically explainable diabetes-risk prediction system** using multiple independent datasets.

Proposed methodology

Instead of evaluating models only within existing datasets, I would:

1. Train Decision Tree, Logistic Regression and Random Forest models on one diabetes dataset.
2. Apply the trained models to a completely independent dataset.
3. Compare:
    - Accuracy
    - Precision
    - Recall
    - F1-score
    - ROC-AUC
    - PR-AUC
    - Sensitivity
    - Specificity
4. Apply SHAP or another explainability method.
5. Evaluate performance separately across demographic groups.
6. Calibrate the predicted probabilities.

Why this is novel?

The original paper demonstrates that performance is **data-driven**, but an extension involving external validation, calibration and subgroup fairness would provide stronger evidence about whether the model can generalize to real clinical environments.

Expected Impact

The experiment could identify whether a model that performs well on one dataset remains reliable on another population.

For example:

```
Training Hospital
        ↓
ML Model
        ↓
External Hospital
        ↓
Performance Evaluation
        ↓
Fairness + Explainability
        ↓
Clinical Decision Support
```

This would move the research closer to practical healthcare deployment.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand